

Code-Layout, Readability and Reusability

Date	12 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation used throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Files organized into templates, static, model, dataset, and app.py.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables clearly represent their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are descriptive and easy to understand.
5	Code Comments	Inline and block comments explain logic	Partial	Important sections are documented.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate folders for UI, dataset, model, and application logic.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed wherever possible.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Function	Python	app.py	High

2	ML Model	Scikit-learn	Prediction Module	High
3	HTML Templates	HTML/CSS	User Interface	High
4	CSS Styles	CSS	All Pages	High
5	Dataset Loader	Pandas	Model Prediction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized
Readability	5	Easy to understand
Reusability	5	Reusable modules
Documentation / Comments	4	Basic comments added
Overall Score	4.8/5	Good quality implementation